

Alice Gao

University of Washington

email: atgao@cs.washington.edu

URL: atgao.github.io

LINKEDIN: linkedin.com/in/atgao/

Areas of interest

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

Education

2023-present

PHD in Computer Science & Engineering, **University of Washington**

2021-2023

MSC in Computer Science & Engineering, **University of Washington**

2017-2021

BS in Computer Science, **Princeton University**

Publications

* denotes equal contribution

CONFERENCE AND JOURNAL PAPERS

- [c.2] **Alice Gao**, Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. *ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*. Montreal, QC, Canada. June 2026. [\[DOI\]](#)
- [c.1] Nino Migineishvili, **Alice Gao**, Adinawa Adjabodjou, Dhanaraj Thaku, René Just, Katharina Reinecke. Regulating AI: Where U.S. State Policy and HCI (Mis)align. *ACM Conference on Human Factors in Computing Systems (CHI '26)*. Barcelona, Spain. April 2026. [\[DOI\]](#)
- [j.1] **Alice Gao**,* Samyukta Jayakumar,* Marcello Maniglia,* Brian Curless, Ira Kemelmacher-Shlizerman, Aaron R. Seitz and Steven M. Seitz. Don't Look at the Camera: Achieving Perceived Eye Contact. *Journal of Vision* 2025; 25(11):8. [\[DOI\]](#)

WORKSHOP PAPERS & POSTERS

- [w.1] K.J. Kevin Feng*, **Alice Gao***, Johanna Suvi Karrass*. Towards Semantically Aware Word Cloud Shape Generation. *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*. Bend, OR. October 2022.

PREPRINTS

- [p.2] **Alice Gao**, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces.

April 2026. *In Submission*. [\[PDF\]](#)

[p.1] Donghoon Shin, **Alice Gao**, Rock Pang, Katharina Reinecke, Emily Tseng. Interrogating Design Homogenization in Web Vibe Coding. March 2026. *In Submission*. [\[PDF\]](#).

Research Experience

- Jan 2024 - Present
Research Assistant, **University of Washington**
Advisor: R. Benjamin Shapiro
- Developed novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing. [p.2]
 - (*Advised w/ Katharina Reinecke*) Developed comparison-based workflow using quantitative metrics for website designs to support UI/UX designers in creating website design for broader audiences.
 - (*Advised by Katharina Reinecke*) Designed a new interventions to reduce AI overreliance during AI-supported writing by up to 30%. [c.2]
- Sep 2021 - Jan 2024
Research Assistant, **GRAIL, University of Washington**
Advisors: Steve Seitz, Brian Curless, Ira Kemelmacher-Shlizerman
Developed machine learning techniques to create a model to redirect attention and eye gaze for a more attentive video conferencing experience. [j.1]
- Sep 2019 - Feb 2021
Undergraduate Research Assistant, **Princeton Vision & Learning Lab, Princeton University**
Advisor: Jia Deng; mentored by Hei Law
Improved associative embedding predictions and grouping from a neural network to provide faster and more accurate predictions for a one-stage object detector in instance segmentation.

Teaching

- University of Washington** (graduate)
- Fall 2026 CSE442: Data Visualization, *Teaching Assistant*
- Spring 2026 CSE446: Machine Learning, *Teaching Assistant*
- Fall 2025 CSE373: Data Structures & Algorithms, *Teaching Assistant*
- Spring 2024 CSE581: Computer Ethics, *Teaching Assistant*
- Fall 2024 CSEP557: Current Trends In Computer Graphics, *Teaching Assistant*
- Winter 2024 CSE581: Computer Ethics, *Teaching Assistant*
- Princeton University** (undergraduate)
- Spring 2020 COS426: Computer Graphics, *Undergraduate Teaching Assistant*
- Fall 2018-Fall 2019 COS226: Algorithms & Data Structures, *Course Grader*

Industry Experience

- Jun-Aug 2020 *Software Engineering Intern in Online Account Opening; Capital One; New York City, New York*
Created new endpoint and unit tests, laid framework for streaming data platform, and launched new security groups.

- Jun-Aug 2019 *AI Design Lab Intern; Tezign; Shanghai, China*
Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs.
- Jun-Aug 2018 *Technical and App Development Intern; Princeton Satellite Systems; Princeton, New Jersey*
Devised beginnings of gameplay for spacecraft simulation iOS app. Created 3D models for Phase II version of Direct Fusion Drive (DFD), a direct-drive, fusion-powered rocket engine.

Service and Leadership

- May 2018 -
Mar 2020 *co-President, Princeton Chinese Students Association*
Started a new Red Envelope fundraiser, acted as a liaison between guest speakers and CSA, participated in creating the first Asian+Students Council, organized and raised funding to host Steven Lim as a guest speaker, started weekly community nights (mahjong nights) that persist until today.

Skills

Languages: Python, R, JavaScript, SQL, TypeScript, C, Go
Frameworks: PyTorch, scikit-learn, Pandas, React, Sveltekit, Node.js, Django/DRF, Flask
Tools: Docker, Apptainer, Figma
Other: Machine Learning, Quantitative and qualitative research methods, Statistical modeling, Experimental design, Semi-structured Interviews

Last updated: September 12, 2026